

CHHAVI NAYYAR

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EDUCATION

BSc (Combined Major in Sciences), University Of British Columbia, Vancouver, BC Sept 2022 – Present

- Focus: Computer Science, Cognitive Systems, Design, Earth and Environmental Science, Life Sciences
- Relevant Coursework: Machine Learning, Algorithms, Data Structures, Web Development, UX Design
- Activities: Design Director @ UBC UX Hub | Member – Women in Computer Science, Girls in STEAM

WORK EXPERIENCE

Machine Learning Engineer (Research), UBC, Vancouver Apr 2025 – Present

- Designed and implemented deep learning models (VTNet) for adaptive educational systems, improving model accuracy by 25% using TensorFlow and eye-tracking data.
- Optimized data preprocessing pipelines and scan path extraction methods for higher model performance and reduced computation time.
- Worked with high-performance computing workflows and Slurm job templates, improving model deployment efficiency by 30%.

Design Assistant (Part-Time), UBC Extended Learning, Vancouver Sept 2024 – Present

- Enhanced LMS course pages by applying user-centred design principles to improve UX and increase user engagement by 15%.
- Conducted A/B testing and collaborated with cross-functional teams to deliver design improvements, improving learner satisfaction by 20%.

Software Engineer (Internship), TechyWeb Solutions, Vancouver Jan 2024 – Mar 2024

- Developed and maintained scalable web applications using React, Node.js, and Next.js, contributing to backend infrastructure using REST APIs and improving performance by 30%.
- Collaborated on the implementation of service-oriented architecture (SOA) to improve modularity and reduce maintenance costs by 25%.

PROJECTS

Your Search Box AI Powered Widget (Personal) | Node.js, React, NLP, AI, Next.js Present

- Developed an NLP-powered personalized search widget, improving accessibility and reducing load times by 25%.
- Integrated voice and visual search features for an enhanced user experience.

Webability Compliance Website (Personal) | CSS, ADA, WCAG Present

- Developed an accessibility compliance tool with features like adjustable fonts, high-contrast modes, and keyboard navigation, improving accessibility by 30%.
- Enhanced website usability for a broad range of users with diverse accessibility needs.

Recommendation System for VR Experiences (Personal) | Python, TensorFlow, Jupyter, Pandas March 2025

- Designed collaborative and content-based filtering algorithms, improving recommendation accuracy by 20% using TensorFlow and applied supervised learning and data analytics to build robust recommendation models.

Autonomous Arduino Robot (Academic) | Arduino, C++, PID, Ultrasonic & IR Sensors April 2025

- Created an autonomous robot with real-time ultrasonic sensor feedback and PID control loops, achieving 98% pathfinding accuracy.
- Implemented algorithmic solutions for real-time decision making and efficient pathfinding.

SKILLS

- **Languages & Frameworks:** TypeScript, JavaScript, Node.js, Java, Python, C++, Express.js, React.js (Hooks), Next.js, Material UI, Redux, Formik, React Hook Form
- **Backend & Cloud:** REST APIs, Firebase, Docker, GitLab CI/CD, Kubernetes, AWS (EC2, S3, Lambda), GCP (basic), Linux
- **Frontend/UI:** HTML5, CSS3, React.js (Hooks), Next.js, Material UI, Redux, Formik, React Hook Form
- **Machine Learning Skills:** TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, NumPy, Jupyter, Time Series Analysis, Large Language Models (LLMs), Supervised & Unsupervised Learning, Reinforcement Learning
- **Testing & Automation:** Jest, Playwright, Mocha, Test-Driven Development, Chrome DevTools
- **Infrastructure & Architecture:** Microservices, API Design, Module Federation, Monorepo, Prometheus
- **Tools:** GitHub, Visual Studio Code, Jira, Slack, Agile/Scrum, Docker, Kubernetes, GitLab